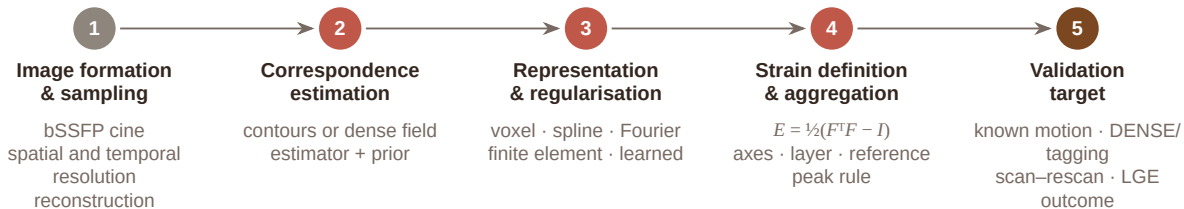


a From cine images to a regional strain report

● observed in cine ● inferred ● reference



b Three failure points, each testable on its own

① Input sufficiency

Do the images contain the cue?
Test: re-image the same motion more finely or with more views

② Estimation

Does the optimiser or network recover it?
Test: vary α (penalty weight) with images fixed

③ Representation capacity

Can the basis express the lesion?
Test: project the known field onto the function space

If a known focal deficit is under-recovered

recovers when α is reduced → **estimation**

still fails after projecting the known field → **representation**

recovers only after re-imaging → **input information**

c Validation must follow the finest claimed spatial and temporal scale; credibility earned upstream does not transfer

→ output feeds the next step

① failure point at step n

Conceptual schematic; no study data.